Trigonometry: sin(x), cos(x), tan(x) in Radians

Questions 1–53

Question 1. Sketch the following graphs for $0 \leq x \leq 2\pi$:

a) $y = \sin x$

b) $y = \cos x$

c) $y = \tan x$

Question 2. Solve the following equations:

a) $\sin\theta = 0, \quad 0 \leq \theta < 2\pi$

b) $\sin\theta = 1, \quad 0 \leq \theta < 2\pi$

c) $\sin\theta = -1, \quad 0 \leq \theta < 2\pi$

Question 3. Solve the following equations:

a) $\cos\theta = 0, \quad 0 \leq \theta < 2\pi$

b) $\cos\theta = 1, \quad 0 \leq \theta < 2\pi$

c) $\cos\theta = -1, \quad 0 \leq \theta < 2\pi$

Question 4. Solve the following equations:

a) $\tan\theta = 0, \quad 0 \leq \theta < 2\pi$

b) $\tan\theta = 1, \quad 0 \leq \theta < 2\pi$

c) $\tan\theta = -1, \quad 0 \leq \theta < 2\pi$

Question 5. Find the values of x for which $y = \sin x$ is a decreasing function, for $0 \leq x < 2\pi$.

Question 6. Find the values of x for which $y = \cos x$ is an increasing function, for $0 \leq x < 2\pi$.

Question 7. Solve the following equations:

a) $\sin\theta = \frac{1}{4}, \quad 0 \leq \theta < 2\pi$

b) $\cos\theta = \frac{3}{4}, \quad 0 \leq \theta < 2\pi$

c) $\tan\theta = \frac{5}{4}, \quad 0 \leq \theta < 2\pi$

Question 8. Solve the following equations:

a) $\sin\theta = 0.9, \quad -2\pi \leq \theta < 0$

b) $\cos\theta = \frac{1}{9}, \quad -2\pi \leq \theta < 0$

c) $\tan\theta = \frac{6}{5}, \quad -2\pi \leq \theta < 0$

Question 9. Solve the following equations:

a) $\sin\theta = -\frac{1}{6}, \quad 0 \leq \theta < 2\pi$

b) $\cos\theta = -\frac{8}{9}, \quad 0 \leq \theta < 2\pi$

c) $\tan\theta = -\frac{1}{3}, \quad 0 \leq \theta < 2\pi$

Question 10. Solve the following equations:

a) $\sin\theta = -0.66, \quad -2\pi \leq \theta < 0$

b) $\cos\theta = -0.55, \quad -2\pi \leq \theta < 0°$

c) $\tan\theta = -8.6, \quad -2\pi \leq \theta < 0$

Question 11. Solve the following equations:

a) $\sin\theta = \frac{4}{9}, \quad -2\pi \leq \theta < 2\pi$

b) $\cos\theta = 0.11, \quad -2\pi \leq \theta < 2\pi$

c) $\tan\theta = 3.2, \quad -2\pi \leq \theta < 2\pi$

d) $\sin\theta = -0.83, \quad -2\pi \leq \theta < 2\pi$

e) $\cos\theta = -\frac{10}{99}, \quad -2\pi \leq \theta < 2\pi$

f) $\tan\theta = -6.6, \quad -2\pi \leq \theta < 2\pi$

Question 12. Find the values of x for which $y = \sin x$ is an increasing function, for $-5\pi \leq x \leq 0$.

Question 13. Find the values of x for which $y = \cos x$ is a decreasing function, for $7\pi \leq x \leq 10\pi$.

Question 14. Solve the following equations:

a) $\sin\theta = \frac{3}{16}, \quad 0 \leq \theta < 4\pi$

b) $\cos\theta = -\frac{5}{11}, \quad 3\pi \leq \theta < 6\pi$

c) $\tan\theta = -0.94, \quad -7\pi \leq \theta < -5\pi$

Question 15. Solve the following equations:

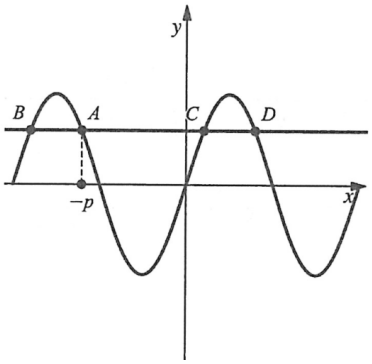
a) $\frac{1}{\sin\theta} = 16.5$ for $0 \leq \theta < \pi$

b) $\frac{14}{\tan\theta} = 19$ for $\pi \leq \theta < 4\pi$

c) $\frac{0.02}{\cos\theta} = 0.3$ for $-\frac{5\pi}{2} \leq \theta < -\frac{3\pi}{2}$

Question 16. A student is solving a problem with a triangle. They correctly use the sine rule to find $\frac{\sin A}{150} = \frac{\sin 1^{c}}{135}$. Find the two possible values of A.

Question 17. The graph below shows the curve $y = \sin x$ for $-2\pi \leq x \leq 2\pi$, and the line $y = k$ where $0 < k < 1$. The point A has x-coordinate $-p$, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.

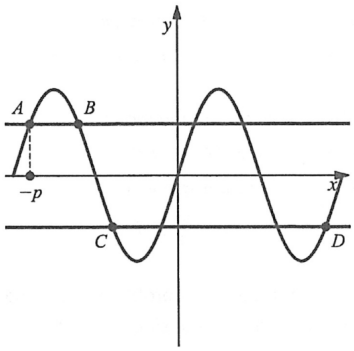


Question 18. It is given that $\sin \frac{\pi}{6} = \frac{1}{2}$.
Explain why $\sin \frac{17\pi}{6} = \frac{1}{2}$.

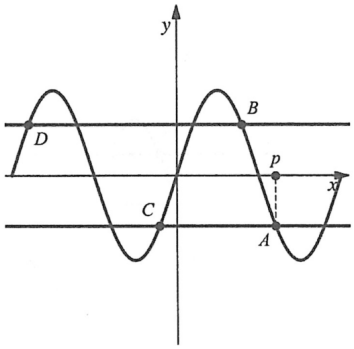
Question 19. It is given that $\cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$.
Explain why $\cos(-\frac{29\pi}{4}) = -\frac{\sqrt{2}}{2}$.

Question 20. It is given that $\tan(-\frac{39\pi}{4}) = 1$.
Explain why $\tan\frac{19\pi}{4} = -1$.

Question 21. The graph below shows the curve $y = \sin x$ for $-2\pi \leq x \leq 2\pi$, and the lines $y = k$ and $y = -k$ where $0 < k < 1$. The point A has x-coordinate $-p$, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



Question 22. The graph below shows the curve $y = \sin x$ for $-2\pi \leq x \leq 2\pi$, and the lines $y = k$ and $y = -k$ where $0 < k < 1$. The point A has x-coordinate p, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



Question 23. Solve the following equations:

a) $\sin^2\theta = \frac{1}{25}$ for $0 \leq \theta < 3\pi$

b) $16\tan^2\theta = 19$ for $-3\pi \leq \theta < -\pi$

c) $25\cos^2\theta = \frac{1}{4}$ for $\pi \leq \theta < 4\pi$

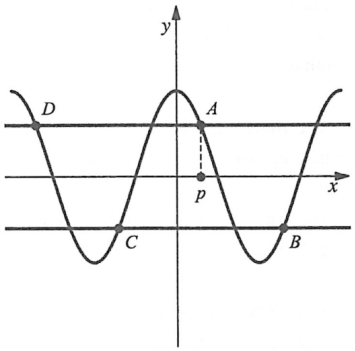
Question 24. Solve the following equations:

a) $\sin^2\theta = 8\sin\theta$ for $0 \leq \theta < 2\pi$

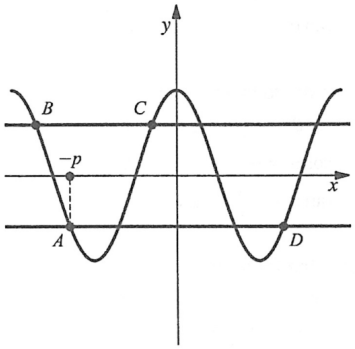
b) $9\tan^2\theta - 5\tan\theta = 0$ for $0 \leq \theta < 2\pi$

c) $-6\cos\theta = 11\cos^2\theta$ for $0 \leq \theta < 2\pi$

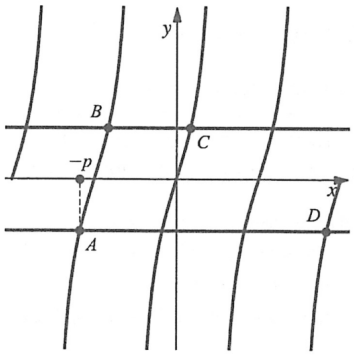
Question 25. The graph below shows the curve $y = \cos x$ for $-2\pi \leq x \leq 2\pi$, and the lines $y = k$ and $y = -k$ where $0 < k < 1$. The point A has x-coordinate p, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



Question 26. The graph below shows the curve $y = \cos x$ for $-2\pi \leq x \leq 2\pi$, and the lines $y = k$ and $y = -k$ where $0 < k < 1$. The point A has x-coordinate $-p$, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



Question 27. The graph below shows the curve $y = \tan x$ for $-2\pi \leq x \leq 2\pi$, and the lines $y = k$ and $y = -k$ where $0 < k < 1$. The point A has x-coordinate $-p$, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



Question 28. Solve the following equations:

a) $5\sin\theta\cos\theta = 3\cos\theta$ for $0 \leq \theta < 2\pi$

b) $\frac{2}{5}\sin\theta\tan\theta + 3\sin\theta = 0$ for $0 \leq \theta < 2\pi$

c) $29\sin\theta - 31\sin\theta\cos\theta = 0$ for $0 \leq \theta < 2\pi$

Question 29. Solve the following equations:

a) $6\sin^2\theta - \sin\theta - 1 = 0$ for $0 \leq \theta < 2\pi$

b) $15\cos^2\theta - 2\cos\theta - 1 = 0$ for $0 \leq \theta < 2\pi$

c) $18\tan^2\theta + 15\tan\theta + 2 = 0$ for $0 \leq \theta < 2\pi$

Question 30. Solve the following equations for $0 \leq \theta < 2\pi$:

a) $32\sin\theta = \frac{9}{\sin\theta} - 12$

b) $54\cos\theta = 51 + \frac{14}{\cos\theta}$

c) $20\tan\theta = 48 + \frac{77}{\tan\theta}$

Question 31. It is given that $f(x) = 5x^2 + 70x + 219$.

a) Write $f(x)$ in the form $a(x + b)^2 + c$, where a, b and c are integers.

b) Write down the minimum point of the curve with equation $y = f(x)$

It is given that $g(\theta) = 5\cos^2\theta - 70\cos\theta + 219$.

c) Using your answers to a) and b), find the minimum value of $g(\theta)$ and the smallest positive value of θ, in radians, for which this minimum occurs.

Question 32. Write down the period of the following functions in radians:

a) $y = \sin x$

b) $y = \cos x$

c) $y = \tan x$

d) $y = \sin 3x$

e) $y = \cos 2x$

f) $y = \tan 6x$

g) $y = \sin \frac{1}{5}x$

h) $y = \cos \frac{8}{3}x$

i) $y = \tan \frac{9}{16}x$

Question 33. State the number of solutions to the following equations. You are not required to solve them.

a) $\sin 2x = 0.7$ for $0 \leq x < 2\pi$

b) $\sin 2x = 0.7$ for $0 \leq x < 3\pi$

c) $\sin 2x = 0.7$ for $-4\pi \leq x < 8\pi$

Question 34. State the number of solutions to the following equations. You are not required to solve them.

a) $\cos 2x = 0.2$ for $-\pi \leq x < \pi$

b) $\cos 3x = 0.2$ for $0 \leq x < 5\pi$

c) $\cos 4x = 0.2$ for $-6\pi \leq x < \frac{3\pi}{2}$

Question 35. State the number of solutions to the following equations. You are not required to solve them.

a) $\tan 2x = 2.1$ for $-2\pi \leq x < 3\pi$

b) $\tan 3x = 2.1$ for $-5\pi \leq x < 10\pi$

c) $\tan 4x = 2.1$ for $-25\pi \leq x < 40\pi$

Question 36. Write down the period of the following functions in radians:

a) $y = \sin(x + \pi)$

b) $y = \cos(x - \frac{2\pi}{3})$

c) $y = \tan(x - 0.1)$

d) $y = \sin(4\pi - x)$

e) $y = \cos(0.9 - x)$

f) $y = \tan(1.3 - x)$

g) $y = \sin(3x + \frac{\pi}{9})$

h) $y = \cos(6x - 0.34)$

i) $y = \tan(12x + \frac{2\pi}{5})$

j) $y = \sin(0.4 - \frac{1}{6}x)$

k) $y = \cos(\frac{0.56 - x}{0.16})$

l) $y = \tan(\frac{5\pi - 15x}{12})$

Question 37. Solve the following equations:

a) $\sin(\theta + 0.32) = 0.45$ for $0 \leq \theta < 2\pi$

b) $\sin(\theta - \frac{8\pi}{9}) = \frac{6}{7}$ for $0 \leq \theta < 2\pi$

c) $\sin(\theta - 1.2) = -0.33$ for $0 \leq \theta < 2\pi$

Question 38. Solve the following equations:

a) $\cos(\theta - 0.97) = 0.2$ for $0 \leq \theta < 2\pi$

b) $\cos(\theta + 2.3) = 0.49$ for $0 \leq \theta < 2\pi$

c) $\cos(\theta - \frac{5\pi}{3}) = -0.89$ for $0 \leq \theta < 2\pi$

Question 39. Solve the following equations:

a) $\tan(\theta + 2.01) = 6$ for $0 \leq \theta < 2\pi$

b) $\tan(\theta - \frac{\pi}{5}) = \frac{3}{8}$ for $0 \leq \theta < 2\pi$

c) $\tan(\theta + 0.99) = -8.2$ for $0 \leq \theta < 2\pi$

Question 40. Find the values of x for which $y = \sin(x - \frac{\pi}{4})$ is a decreasing function, for $0 \leq x \leq 2\pi$.

Question 41. Solve the following equations:

a) $\sin 2\theta = \frac{\sqrt{2}}{2}$ for $0 \leq \theta < 2\pi$

b) $\cos 2\theta = \frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$

c) $\tan 2\theta = \sqrt{3}$ for $0 \leq \theta < 2\pi$

Question 42. Solve the following equations:

a) $\sin 3\theta = 0.76$ for $0 \leq \theta < 2\pi$

b) $\cos 3\theta = -0.95$ for $0 \leq \theta < 2\pi$

c) $\tan 3\theta = -0.32$ for $0 \leq \theta < 2\pi$

Question 43. Solve the following equations:

a) $\sin \frac{1}{2}\theta = -0.28$ for $0 \leq \theta < 4\pi$

b) $\cos \frac{1}{5}\theta = \frac{2}{9}$ for $0 \leq \theta < 10\pi$

c) $\tan \frac{7}{5}\theta = -2.4$ for $0 \leq \theta < \pi$

Question 44. Solve the following equations:

a) $\sin(3\theta - 0.31) = \frac{2}{7}$ for $0 \leq \theta < \pi$

b) $\cos(4\theta + 0.97) = \frac{6}{11}$ for $0 \leq \theta < \pi$

c) $\tan(5\theta + 1.45) = \frac{7}{8}$ for $0 \leq \theta < \frac{\pi}{2}$

Question 45. Solve the following equations:

a) $\sin(5\theta + 3.2) = -0.16$ for $-\pi \leq \theta < 0$

b) $\cos(7\theta - 4.2) = -0.12$ for $\pi \leq \theta < \frac{3\pi}{2}$

c) $\tan(3\theta - 6.4) = -12$ for $2\pi \leq \theta < 4\pi$

Question 46. Find the values of x for which $y = \cos 3x$ is a decreasing function, for $0 \leq x \leq \pi$.

Question 47. Find the values of x for which $y = \sin \frac{1}{4}x$ is an increasing function, for $0 \leq x \leq 10\pi$.

Question 48. Solve the following equations:

a) $\sin(\frac{\pi}{8} - 2\theta) = \frac{\sqrt{2}}{2}$ for $0 \leq \theta < 2\pi$

b) $\cos(\frac{3\pi}{5} - 3\theta) = \frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$

c) $\tan(\frac{3\pi}{10} - 4\theta) = 1$ for $0 \leq \theta < 2\pi$

Question 49. Solve the following equations:

a) $\sin(1.2 - 4\theta) = -0.07$ for $0 \leq \theta < 2$

b) $\cos(2.1 - 5\theta) = 0.93$ for $0 \leq \theta < 1.5$

c) $\tan(4.3 - 7\theta) = -0.16$ for $0 \leq \theta < 1$

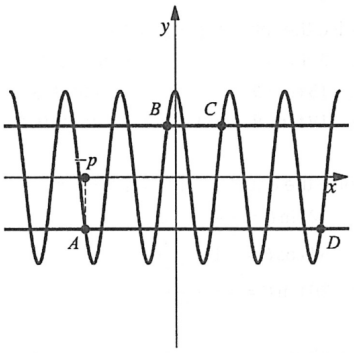
Question 50. Solve the following equations:

a) $\sin^2 4\theta - \frac{1}{25} = 0$ for $-1.4 \leq \theta < 1.4$

b) $\cos^2 3\theta - \frac{1}{81} = 0$ for $-2 \leq \theta < 2$

c) $\tan^2 9\theta - 169 = 0$ for $-0.5 \leq \theta < 0.5$

Question 51. The graph below shows the curve $y = \cos 3x$ for $-2\pi \leq x \leq 2\pi$, and the lines $y = k$ and $y = -k$ where $0 < k < 1$. The point A has x-coordinate $-p$, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



Question 52. Solve the following equations:

a) $\sin(\theta - \frac{\pi}{3}) = \sin\frac{\pi}{6}$ for $0 \leq \theta < 2\pi$

b) $\sin(\theta + \frac{2\pi}{5}) = \sin\frac{11\pi}{10}$ for $0 \leq \theta < 2\pi$

c) $\cos(\theta + \frac{3\pi}{7}) = \cos\frac{4\pi}{3}$ for $0 \leq \theta < 2\pi$

d) $\tan(\theta - \frac{5\pi}{8}) = \tan\frac{\pi}{16}$ for $0 \leq \theta < 2\pi$

Question 53. Solve the following equations:

a) $\sin(\frac{\pi}{3} - \theta) = \sin\frac{46\pi}{9}$ for $0 \leq \theta < 2\pi$

b) $\cos(2\theta - \frac{5\pi}{6}) = \cos\frac{11\pi}{12}$ for $0 \leq \theta < 2\pi$

c) $\tan(\frac{7\pi}{6} - 4\theta) = \tan\frac{8\pi}{3}$ for $0 \leq \theta < 2\pi$